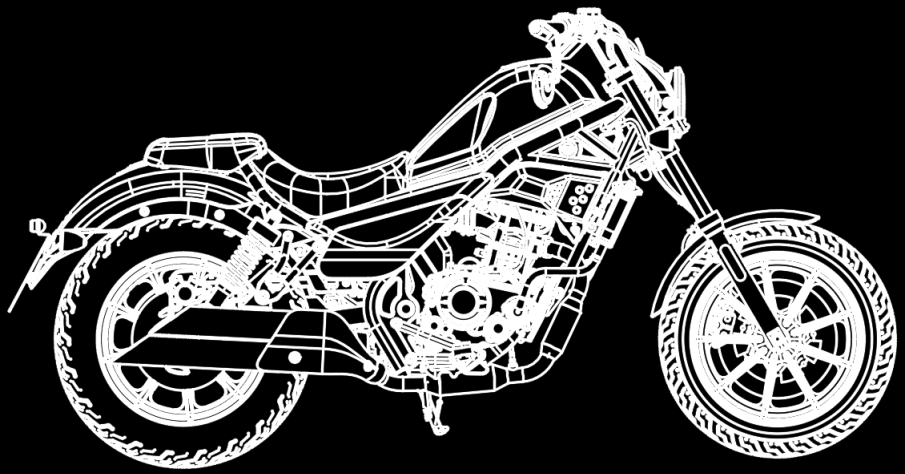


Owner's Manual

MITT125

Thank you for choosing a MITT motorcycle. Our motorcycles are crafted with meticulous design, rigorous testing, and precision manufacturing. We sincerely welcome you to the world of motorcycle riding.

This manual contains important information and instructions, including fundamental operations, inspection procedures, and maintenance methods. Please read it carefully before operating the motorcycle. Proper use and maintenance will help keep your vehicle in optimal condition and extend its service life. If you have any questions regarding operation or maintenance, please contact Marshalmoto customer service.

This vehicle is designed and manufactured in strict accordance with national standards. Through careful engineering and precision production, we ensure excellent overall performance. Please make sure to use only genuine original parts. Once you are familiar with the instructions and recommendations in this manual, you will find that riding a motorcycle is an exciting and exhilarating experience, allowing you to truly enjoy the pleasure of motorcycling.

We wish you comfort and enjoyment throughout your journey with your motorcycle.

⚠ Safety Warning Symbol: Indicates that you need to pay attention, as your personal safety may be at risk.

⚠ WARNING Warning Symbol: Failure to follow the warning instructions may result in serious injury or even death.

CAUTION Caution Symbol: Indicates that certain actions must be taken to avoid damage to the vehicle.

NOTE: Note Symbol: Provides key information that is important for proper operation or maintenance.

NOTE:

● This manual should be regarded as a permanent part of the motorcycle. If the vehicle is transferred to another owner, please make sure this manual is included with the vehicle.
● Our company is committed to continual improvement in product design and quality. Therefore, the data, descriptions, specifications, and images contained in this manual represent the latest information available at the time of printing. They may differ slightly from your actual motorcycle. If you have any questions regarding the contents of this manual, please contact your local Marshalmoto dealer. Marshalmoto reserves the right to make changes at any time without prior notice.
● The copyright of this manual is owned by Marshalmoto. No part of this manual may be reproduced or copied without the written permission of our company.

This manual applies to the MITT-125 model. It provides detailed instructions on the basic operation and maintenance of the vehicle. Please read it carefully before riding. Proper use, care, and maintenance will effectively ensure riding safety, reduce failures, and keep the motorcycle in optimal condition.

Thank you for choosing and using the Marshalmoto range of products!

⚠ WARNING

●Please read this manual carefully and do not operate your motorcycle until you are riding in a safe area and have acquired the necessary riding skills. Please perform regular inspections and maintenance, and always maintain good riding habits to ensure safe control of the vehicle.

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Motorcycles are classified as motor vehicles.
Operating this type of vehicle requires proper riding skillsand basic technical knowledge.
Dealers should inform riders of the following:

Riders should be aware of :

- Becoming fully familiar with the operation and control of all components of the motorcycle.
- Paying attention to the warnings and maintenance information provided in this manual.
- Riders must pass the relevant examinations administered by the traffic management authorities and obtain a valid motorcycle driving license before operating the vehicle. Riding without a license is strictly prohibited.
- By reading this owner ' s manual, riders can understand basic information about the motorcycle and important precautions for proper use.

Safe Riding

● Before operating the motorcycle, carefully read the “Pre-Ride Inspection” section and conduct the necessary checks. This helps prevent accidents and ensures safe riding.

● This motorcycle is designed to carry only the rider and one passenger—no more than two persons in total.

● Due to their smaller size, motorcycles are often not easily noticed by car drivers. Many motorcycle–car collisions occur simply because the driver of the car did not see the motorcycle. Increasing your visibility can greatly reduce such accidents.

Therefore:

● Wear bright, noticeable clothing. The rider must wear proper motorcycle riding gear and protective equipment, while the passenger should dress neatly and securely.

● Pay extra attention when riding through intersections or high-risk areas, and make sure you are clearly visible to other drivers.

● Always keep yourself within the visible range of other drivers, and avoid staying in their blind spots.

● Maintain a safe distance from other motor vehicles. Use turn signals, the horn, and other indicators correctly. Never compete for lane position, and always follow traffic regulations.



Properly Dressed



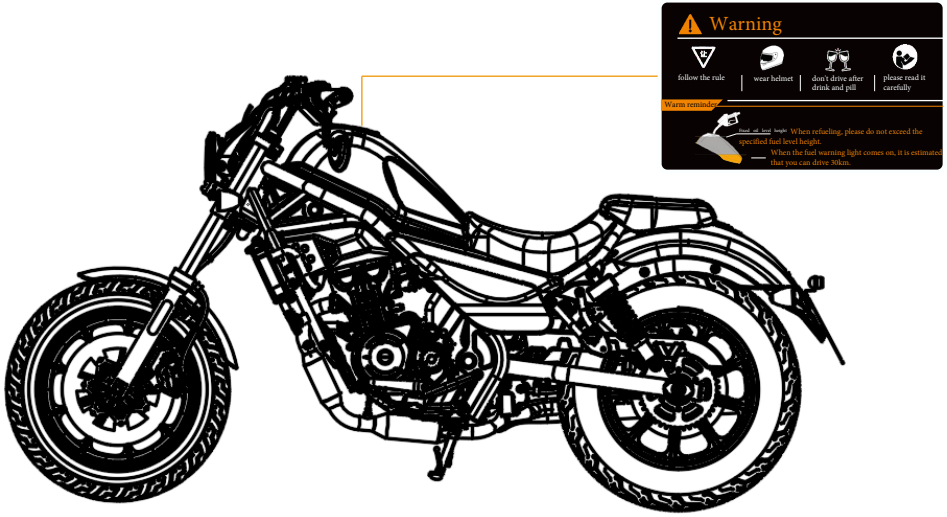
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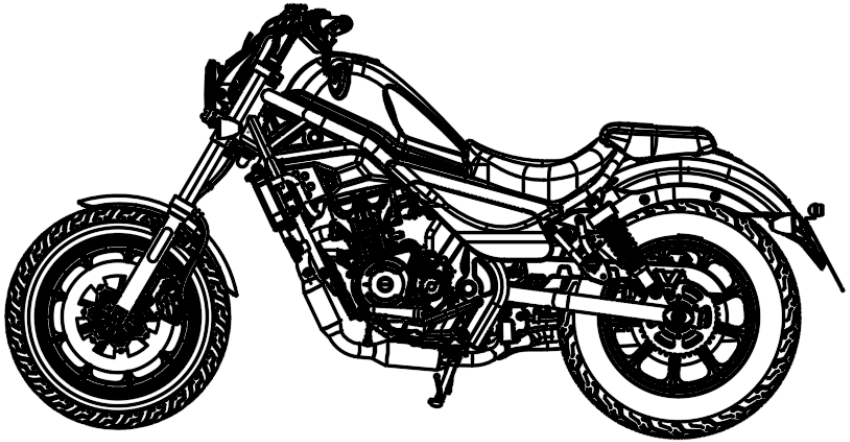
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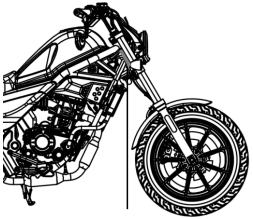
Important Vehicle Label Locations

Please read the label information shown below carefully before operating the motorcycle.

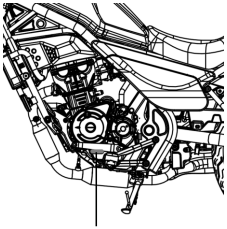


Important Vehicle Label Locations

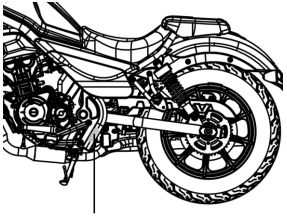




Vehicle Identification Number (VIN) is stamped on the right side of the head tube.

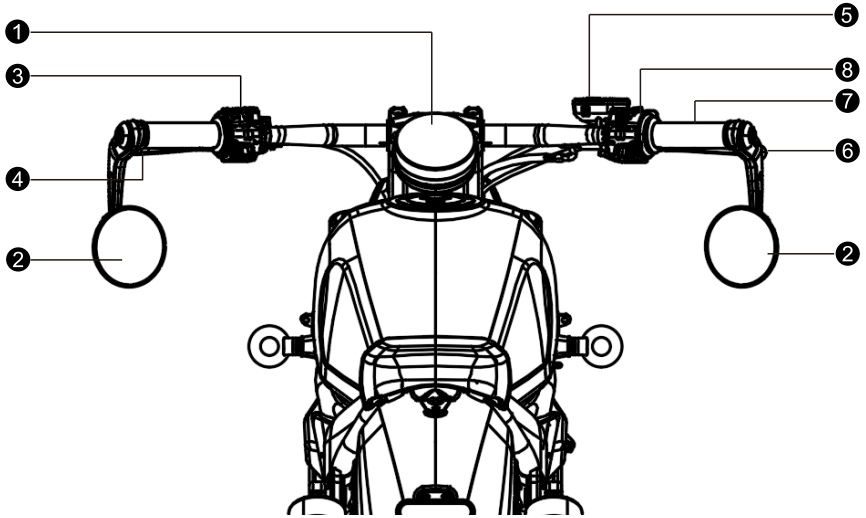


Engine number is stamped on the lower part of the left crankcase.



Vehicle information label is located on the lower left side of the frame.

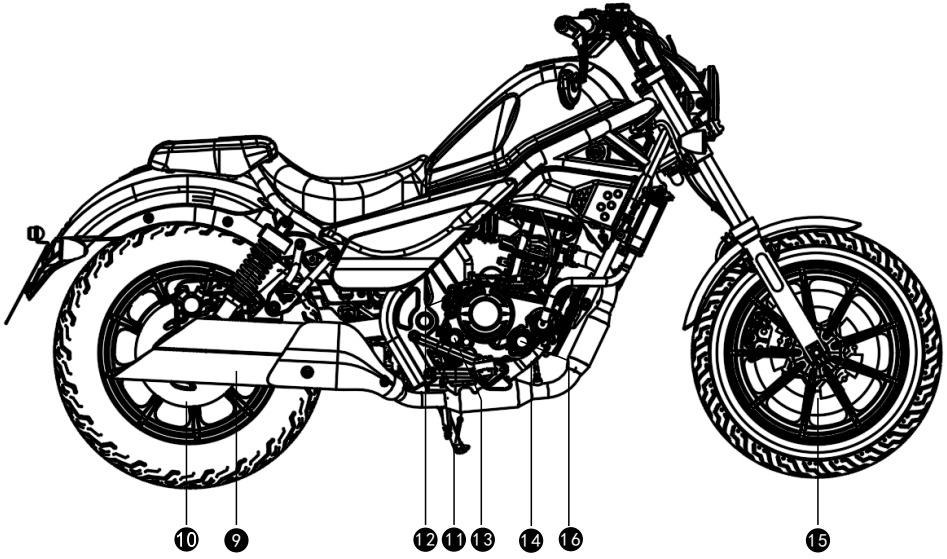
Controls and Instruments



❶ LCD Display Meter
❷ Rearview Mirrors
❸ clutch Lever
❹ Left Handlebar Switch
❺ Front Brake Fluid Reservoir
❻ Front Brake Lever
❼ Throttle Grip
❽ Right Handlebar Switch

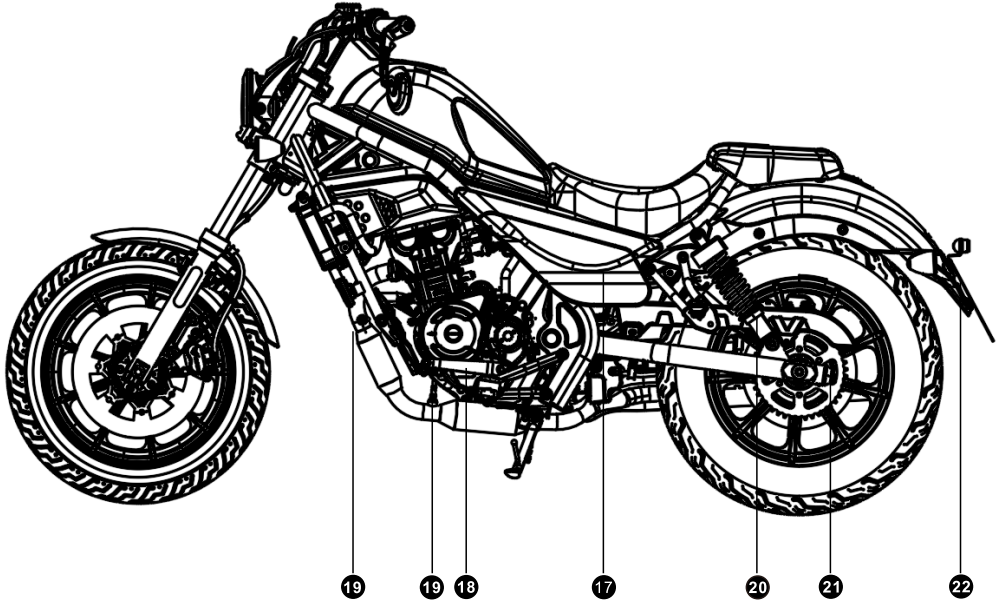
General Components Description

Right Side of Vehicle



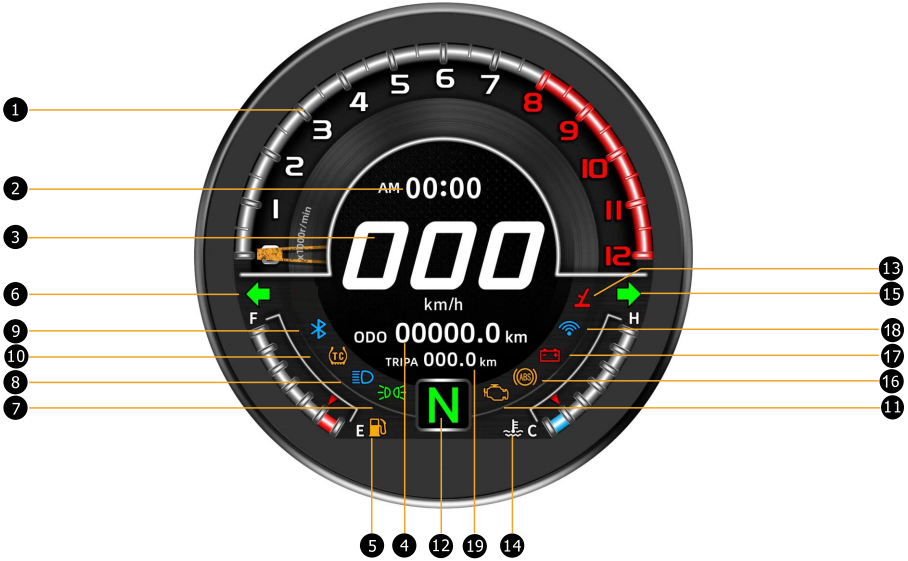
❾ Muffler
❿ Rear Brake Disc
⓫ Side Stand
⓬ Engine Oil Filler Cap
⓭ Engine Oil Level Inspection Window
⓮ Rear Brake Pedal
⓯ Front Brake Disc
⓰ Oil Filter

Left side of the vehicle



17 Air Filter
18 Gear Shift Pedal
19 Oxygen Sensor
20 Rear Shock Absorber
21 Chain Adjuster
22 License Plate Bracket

Instrument Panel



1 Tachometer
2 Clock Display
3 Speedometer
4 Odometer
5 Fuel Gauge
6 Left Turn Indicator
7 Position Light Indicator
8 High Beam Indicator
9 Bluetooth Indicator
10 TCS Indicator
11 Injection Fault Indicator
12 Gear Position Indicator
13 Side Stand Indicator
14 Water Temperature Indicator
15 Right Turn Indicator
16 ABS Fault Indicator
17 Power Fault Indicator
18 WiFi Connection Indicator
19 Trip Meter

⚠ WARNING

● Do not clean the instrument panel with a high-pressure water jet.Never wipe the display with cloths soaked in gasoline, kerosene, alcohol, brake fluid, or other organic solvents, as these may cause cracking or discoloration of the instrument surface.

1. Tachometer: Displays the engine speed (RPM).
2. Clock Display: Shows the current time.
3. Speedometer: Displays the current vehiclea speed.
4. Odometer (ODO): Displays the total accumulated distance traveled by the vehicle.
5. Fuel Gauge Shows the remaining fuel in the tank. When the fuel warning indicator begins to flash, refuel as soon as possible.

CAUTION

● Always keep an adequate amount of fuel in the tank. Running out of fuel may cause the fuel pump to operate without lubrication, which can lead to damage.

6. Left Turn Signal Indicator: Lights when the left turn signal is activated.
7. Position Light Indicator: Lights when the position lights are turned on.
8. High Beam Indicator: Lights when the high beam is switched on.
9. Bluetooth Indicator: Lights when Bluetooth is activated and connected.
10. TCS Indicator (Traction Control System)When TCS is ON, the indicator lights normally. When TCS is OFF, the indicator lights with a “/” symbol to show the system is disabled.

The Traction Control System monitors rear-wheel slip during acceleration. When excessive slip is detected, the system reduces engine torque through ignition retard or fuel cut-off to maintain traction, acceleration, and vehicle stability.

TCS Activation Conditions: When the front wheel speed is higher than 1 km/h, the system continuously monitors the wheel speed difference.If the rear wheel speed exceeds the front wheel speed by 15%, TCS intervenes and attempts to maintain the difference within around 10% to ensure stable acceleration.

Escape Mode (if equipped): When the front wheel speed is below 1 km/h and the rear wheel speed exceeds 15 km/h, TCS activates to maintain the rear wheel speed at 10 – 15 km/h, providing optimal traction.If TCS remains active for more than 5 seconds while the front wheel is stationary, the system determines the vehicle is stuck and automatically disables TCS. Power cycling (turning ignition OFF then ON again) is required to restore TCS function.

11. Fuel Injection System Fault Indicator: The indicator lights when the ignition is turned on and turns off after the engine starts. If it remains lit after the engine starts, it indicates a fault in the EFI system. Do not continue riding. Please contact Marshalmoto after-sales service.
12. Gear Position Indicator: Shows the current transmission gear selected.

13 Side Stand Indicator: The indicator lights when the side stand is lowered. When the side stand is down, the engine will shut off, and the motorcycle cannot be started unless the transmission is in neutral. When the side stand is raised, the motorcycle can be started in neutral, but gear shifting is disabled.

14 Water Temperature Display (Cooling System Temperature): Displays the real-time temperature of the coolant in the radiator.

CAUTION

● The coolant temperature will vary depending on weather conditions and engine load. If the water temperature warning indicator begins to flash, stop the motorcycle and allow the engine to cool down.

15 Right Turn Signal Indicator: Lights when the right turn signal is activated.

16 ABS Fault Indicator: The ABS indicator lights when the ignition is switched on. It should automatically turn off after the motorcycle reaches 10 km/h. If it does not turn off, there may be a malfunction—do not continue riding. Please contact a Marshalmoto service center.

⚠ WARNING

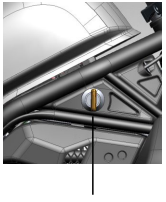
The ABS may be faulty if: :

● The ABS lamp does not light when the ignition is turned on.
● The ABS lamp flashes while riding.
If either condition occurs, visit an authorized Marshalmoto dealer or service station.

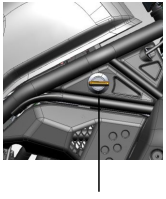
17 Power Fault Indicator: The power fault indicator lights when the ignition is switched on if a system failure or low battery condition is detected. Do not continue riding. Please contact a Marshalmoto service center.

18 WiFi & Bluetooth Indicators: When the mobile phone screen-mirroring function is successfully connected, both the WiFi and Bluetooth indicators illuminate.

19 Trip Meter (TRIP): The trip meter displays short-distance mileage up to 999 km, after which it resets automatically.
● Press the Confirm button briefly to reset the current trip mileage.
● Press the Return button briefly to switch between TRIP A / TRIP B before performing the reset operation.



Turn the ignition key clockwise to the ON position to power on the electrical system.



Turn the ignition key counterclockwise to the OFF position to cut off the electrical system.

Main Power ON

When the main power is switched on, all electrical circuits and components become active, including the instrument Illumination, tail lamp, auxiliary lights, and position lights. At this point, the engine can be started, and the key cannot be removed.

NOTE:

●When the main power is switched on, the headlight (driving light) will automatically turn on, even if the engine is idling. This feature remains active unless the main power switch is turned to an angled intermediate position.

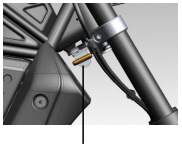
Main Power OFF

All electrical circuits are shut down.

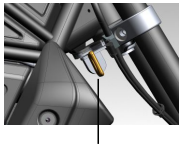
Steering Lock

When the handlebars are turned to the left and the key is pressed down and turned to the LOCK position, the steering mechanism will be locked and the motorcycle cannot be moved.

Steering Lock Operation



Turn the key clockwise to engage the steering lock.

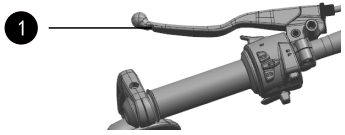


Turn the key counterclockwise to release the steering lock.

⚠ WARNING

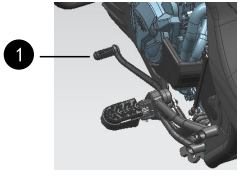
●Do NOT rotate the main power switch to the OFF position while the motorcycle is moving. Turning off the ignition while riding will immediately shut down the electrical system, which may cause loss of control and result in an accident.

Clutch Lever



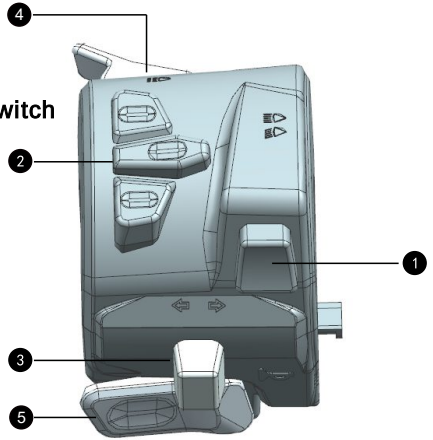
When starting the engine or shifting gears, squeeze the clutch lever to disengage the clutch plates, thereby cutting off power transmission.

Gear Shift Pedal



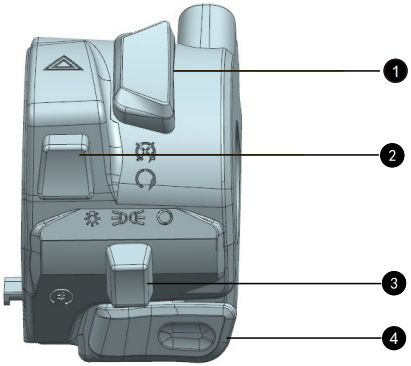
● This motorcycle is equipped with a six-speed transmission. Press downward or lift upward on the gear shift pedal to change gears; the pedal will automatically return to its original position after shifting. Before shifting to a lower gear, please reduce vehicle speed or increase engine RPM. Before shifting to a higher gear, please increase vehicle speed or reduce engine RPM. Doing so helps prevent unnecessary wear to the drivetrain components and rear wheel.

Left Handlebar Switch



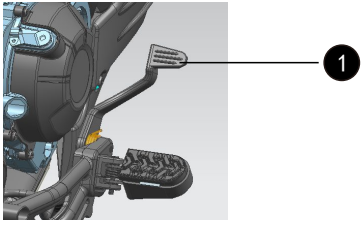
1. High/Low Beam Switch
2. Instrument Operation Switch
3. Turn Signal Switch
4. Passing Light Switch
5. Horn Button

Right Handlebar Switch



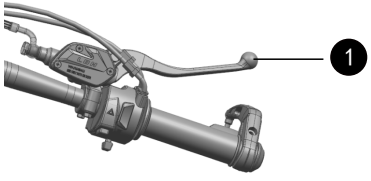
1. Kill Switch (Engine Stop Switch)
2. Hazard Light Switch
3. Headlight Switch
4. Ignition / Start Switch

Rear Brake Pedal



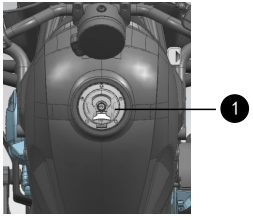
The rear brake pedal is located on the right side of the motorcycle, beside the engine. Press the pedal to operate the rear brake. The rear brake pedal is equipped with an adjustment bolt, which allows you to set the free play of the pedal. Adjust the bolt to maintain the proper free play.

Front Brake Lever



The front brake lever is located on the right side of the handlebar.
Pull the lever to operate the front brake.

Fuel Tank Cap



To open the fuel tank cap: Press the cap downward and turn it counterclockwise until it releases. The cap will pop up automatically.
To close the fuel tank cap: Insert the cap into the filler opening, press it downward, and turn it clockwise to its original locked position until it clicks.

Fuel Tank



Do not fill fuel above the specified maximum level. When the low fuel warning light turns on, the remaining riding range is approximately 20 km.

⚠ WARNING

- Do not fill fuel above the lower edge of the filler neck. Overfilling may cause fuel to overflow when it expands due to heat.
- Avoid spilling fuel onto the engine when it is hot.

CAUTION

●If fuel is spilled, clean it immediately with a dry and soft cloth.
Fuel may stain or damage painted surfaces and plastic components.

NOTE：

Fuel Tank Capacity: 14.0 L
Recommended Fuel: 92 RON unleaded gasoline or higher
This motorcycle is designed to use 92 RON or higher unleaded fuel.
Continuous use of lower-grade gasoline may cause knocking and shorten engine life.

Catalytic Converter

This motorcycle is equipped with an exhaust muffler containing a catalytic converter to reduce emissions.

⚠ WARNING

●The exhaust system becomes extremely hot during engine operation.
When performing maintenance near the muffler or catalytic converter, ensure the system has fully cooled down to avoid burns.

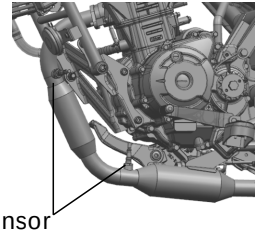
CAUTION

To prevent catalytic converter damage or malfunction, please observe the following:

●Use unleaded gasoline only. Leaded fuel will permanently damage the catalytic converter.

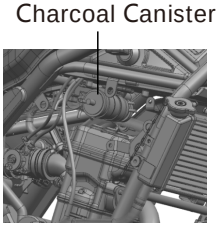
●Do not operate the motorcycle with misfiring or incomplete combustion, as unburned fuel entering the exhaust system may ignite and damage the converter. Avoid parking the motorcycle over dry grass or flammable materials when the engine is hot.Do not idle the engine for prolonged periods.

a
Oxygen Sensor



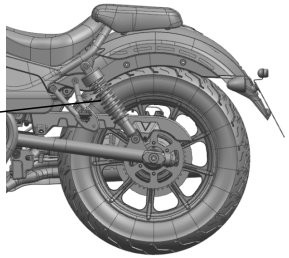
This motorcycle is equipped with an oxygen sensor, which helps reduce exhaust emissions by monitoring the air – fuel mixture.

Charcoal Canister



The motorcycle is equipped with an activated carbon canister to prevent fuel vapors from being released into the atmosphere. Inspect the canister and its connecting hoses at the specified intervals. Check whether the canister and hoses are cracked or damaged. Replace if necessary. Ensure all hose connections are secure after inspection .

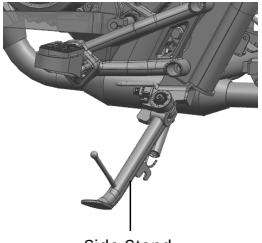
Rear Shock Absorber Assembly



The rear shock absorber can be adjusted according to different road conditions and riding preferences.

Adjustment Method: Loosen the locknut on the preload adjuster first. Rotate the adjuster clockwise to increase preload and stiffen the suspension. Rotate counterclockwise to decrease preload and soften the suspension. Adjust the preload according to rider weight and riding requirements, then retighten the locknut.

Side Stand



WARNING

- Do not ride the motorcycle when the side stand is lowered or cannot be fully retracted. Riding with the side stand down may cause it to contact the ground, resulting in loss of control .

Fuel Pump

This motorcycle is equipped with an electric fuel pump. When the ignition switch is turned ON, the fuel pump operates for 2 – 3 seconds and then stops. After the engine starts, he fuel pump continues to operate. Whe n the fuel is depleted, the fuel pump will run dry, which may cause damage. Please monitor the fuel gauge and refuel befor e the tank becomes empty.

CAUTION

● Use 92 RON or higher unleaded gasoline only. Poor-quality or contaminated fuel may damage the electronic fuel injection system.

CAUTION

● Ensure that there is sufficient fuel in the tank before the fuel pump operates.
Do not run the fuel pump with an empty tank. The fuel pump is designed only for gasoline and must not be run dry.

You must carefully check the following items before riding.

Failure to properly inspect and maintain the motorcycle before riding increases the risk of an accident and may lead to vehicle damage. Before each ride, you must inspect the motorcycle to ensure that it is in safe operating condition. Refer to the Inspection and Maintenance section of this manual for detailed procedures. Using incorrect tires, improper riding techniques, or incorrect tire pressure may result in loss of vehicle control. Always use tires of the specified size and specifications listed in this manual. Maintain proper tire pressure as required in the Inspection and Maintenance section.

1. Engine Oil Level: Check the engine oil level and add oil if necessary.
2. Fuel Leaks: Inspect the vehicle for any signs of fuel leakage.
3. Front and Rear Brakes: Check brake operation. If the free play is too large or too small, adjust accordingly. Ensure there is no dragging, locking, or loss of braking force.
4. Tires: Check tire pressure and tire wear. Inflate or repair as needed.
5. Throttle Operation: Check throttle grip movement, cable connections, and free play. Adjust or replace if necessary.
6. Lights and Signals: Check the headlight, tail light, turn signals, and horn for proper function. Repair or adjust if necessary.
7. Drive Chain: Inspect chain tension and lubrication. Replace if severely worn or damaged.
8. Steering System: Check steering flexibility and stability. Adjust if necessary.
9. Engine Condition: Ensure there are no abnormal engine noises or irregularities.
10. Instruments: Verify that all indicator lights and the instrument panel operate normally. Repair if necessary.
11. Fasteners: Check that all bolts, nuts, and fasteners are tightened properly. Tighten if necessary.
12. Side Stand Cut-Off Switch: Check the ignition cut-off system triggered by the side stand. If it malfunctions, contact a Marshalmoto dealer or service center.

NOTE：

You must never ignore the importance of these inspections. All inspections and any necessary maintenance must be completed before riding.

⚠ WARNING

●Performing inspections while the engine is running can be extremely dangerous. If your hands, clothing, or tools become caught in moving parts, severe injury may occur. Except when checking the engine stop switch or throttle operation (which require the engine to run), turn off the engine before performing all other inspections.

⚠ WARNING

●If you are riding this type of motorcycle for the first time, we recommend practicing in a non-public area until you become fully familiar with its controls and handling characteristics.
●Riding with one hand is extremely dangerous. Always keep both hands on the handlebars and both feet on the footrests. Never release both hands from the handlebars under any circumstances.
●Reduce your speed to a safe level before entering a turn.
●On wet or slippery surfaces, tire traction is reduced, which decreases braking and cornering performance. Always slow down in advance.
●Crosswinds often occur at tunnel exits, in valleys, or when large vehicles overtake you from behind. Stay calm, hold the handlebars firmly, and reduce speed when necessary.
●Obey all traffic regulations and speed limits.

Engine Start

When the motorcycle is powered ON, check that the engine stop switch is in the "RUN/ON (0)" position.

⚠ WARNING

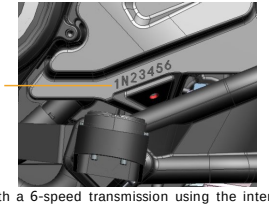
●Develop the habit of starting the engine with the transmission in neutral, the throttle fully closed, and the clutch lever pulled in. This prevents the motorcycle from lurching forward in case of incorrect gear engagement.The clutch lever must be pulled in when starting the engine. If the motorcycle is not in neutral, the engine will not start.Do not forget to retract the side stand before riding. Do not start the engine when fuel or engine oil is insufficient.

⚠ WARNING

The higher the speed, the longer the braking distance. Always maintain a safe following distance from the vehicle ahead. Using only the front brake or only the rear brake is dangerous and may cause the motorcycle to skid and lose control. Always use both brakes together

Gear Shift Pedal Operation

Shifting Pattern



This motorcycle is equipped with a 6-speed transmission using the international shift pattern (one gear down, the rest up). Press the gear shift pedal down to select 1st gear, and lift the pedal up to shift to 2nd gear and above. Shift up or down through the gears as shown in the illustration and always select an appropriate gear for the riding conditions.

CAUTION

●When the transmission is in neutral and the neutral indicat or light is ON, confirm that the transmission i s truly in neutral by slowly releasing the clutc h lever.

Recommended Engine Oil

Engine oil is a critical factor affecting engine performance and service life.Always use the specified oil and never substitute poor-quality engine oil, gear oil, vegetable oil, or any other non-approved lubricant. Use engine oil meeting API service classification SG or higher, with a viscosity grade of: 10W/40 for normal ambient temperatures 0W/30 for cold winter conditionsIf other oil grades are used, their quality level must be SG or higher, and viscosity should be selected according to regional climate and temperature.When changing the engine oil, completely drain the old oil from the crankcase, then refill with the specified quantity of fresh oil.

NOTE:

●Do not dispose of used engine oil directly into the environment. Collect and deliver used oil to an appropriate recycling or disposal facility to avoid environmental pollution.

Tire Pressure & Load

●Correct tire pressure and proper tire load are essential for safe riding. Overloading may cause tire failure and result in loss of vehicle control. Always check the tire ressure before each use to ensure it is within the correct range. Tire pressure must be inspected and adjusted before riding, as tire temperature increases during riding and causes the pressure to rise.

●Insufficient tire pressure may cause difficult handling and accelerated tire wear. Excessive tire pressure reduces the tire ' s contact area with the road, increasing the risk of skidding and loss of control.The recommended tire pressure under normal ambient temperature is 250 kPa.

⚠ WARNING

Failure to observe the following precautions may result in tire failure and an accident.
Tires are the only contact between the motorcycle and the road and are extremely important. Always follow these rules:
●Inspect tire condition and pressure before each ride; adjust tire pressure as required and avoid overloading the motorcycle.
●Replace the tire when tread wear reaches the limit, or if cracks, cuts, or other damage are found on the tire surface. Always use tires of the specified size and specification given in this manual.
●Read this section of the owner's manual carefully.
New tires require a break-in period; improper break-in may cause the tires to slip and result in loss of control. Ride extra carefully when new tires are fitted.

WARNING

● Check the tire pressure regularly, and the tire pressure is kept within the recommended range. When you feel that the tire pressure drops, check whether there are nails, small holes, damage to the rim side, and whether the valve core is leaking.

Tire Storage

When the m otorcyc le is not in use for a period of ti me, the tire pressure needs to be adjusted to the specified air pressure. Tires are semi-hot melt formula rubber products, not suitable in cold outdoor weather, otherwise it will cause the product to freeze crack, pl ease store in a tem perature bal anced place or indoors.

Tire Condition & Specifications

Incorrect tyre condition and incorrect tyre specifications can affect the performance of a motorcycle. A broken scratch on the tire can cause the tire to malfunction and cause the vehicle to lose control . Excessive wear of tires can cause tire punctures and cause the vehicle to lose control. Tire wear also affects the shape of the tire and changes the operating performance of the tire. Check the condition and air pressure of the tyre befor e each use. If there are many visible injuries on the tire, such as breakage, scratches, or if the tyre wears out to the limit, the tire must be replaced.

NOTE:

When changing tires, determine that the size and model of the replacement tire should conform to the contents of the list below. If you change the tires of different sizes or different models, it will affect the handling performance of the motorcycle and may cause the motorcycle to lose control.

Problems may occur with tires other than the standard, and we sincerely recommend that you choose standard tires..

Front wheel	Rear wheel
120/80-16	150/80-16

Engine Break-In

Although the engine parts are selected from the best materials and manufactured according to advanced standards, it is necessary for the new car to run in all moving parts before the engine runs at full load, and the performance and reliability of the engine in the future depends on the strict and m eticu lous running -in and maintenance at the beginning of use. Generally, after the engine is started, it runs at a low speed and fully warms up, and then starts . Drive at a low speed for 1~2 km before turning into normal driving. The general rules are as follows:

Mileage (km)	Throttle opening (%)	Rotate speed (r/min)
Grind in 1000km	≤30%	≤4500r/min
More than 1000km		≤9000r/min

NOTE:

● For new cars, please go to the service outlet of Xiangshuai Heavy Machinery within the designated kilometers for maintenance and replace the oil and oil filter element.

First maintenance	Second maintenance	Third maintenance	Repeat maintenance
300km	800km	1500km	4000km

NOTE:

● After running-in, a maintenance should be carried out, all fasteners should be tightened, and the parts should be readjusted to ensure that the motorcycle is in the best condition.

● If any problem occurs during engine running-in, please contact Xiangshuai heavy machinery motorcycle distributor or service outlet immediately.

WARNING

● When the vehicle is just turned off, since the engine and exhaust pipes will be very hot, please park the vehicle in a place that is not accessible to pedestrians and children to avoid burns.

● Please do not park the car on the slope or soft ground, otherwise the locomotive will roll over, increasing the risk of fuel leakage and fire.

● Do not park locomotives near hay or other flammable materials.

Fuel Injection System Precautions

●Before installing the battery on a new vehicle, check that all fuel-injection wiring harnesses and connectors are securely fastened, including proper installation of the oxygen sensor, and confirm that fuel has been added to the tank.
●When installing the battery, use proper tools to securely fasten the positive and negative cables to the respective battery terminals. Do not tighten the terminals by hand.
●Keep at least 3 liters of fuel in the tank, otherwise the fuel-injection system may not operate correctly. Refill promptly when the fuel gauge shows one bar or below.
●In cases such as battery reinstallation, sudden power loss during riding or starting, battery sleep-wake cycle, abnormal idling, or reconnecting the fuse, perform a reset of certain fuel-injection hardware.

●If the motorcycle has been left unused for an extended period, initial startup may be difficult. Turn the throttle to 1/8 open and then start the engine. If the engine still fails to start after several attempts, the cylinder may be flooded. Perform the deflooding procedure: remove the spark plug, clean it, and reinstall it.
●If the battery voltage indicator is flashing, the battery voltage is too low. Charge the battery immediately. Low voltage may cause the fuel-injection system to malfunction, resulting in hard starting, failure to start, or poor power performance.

⚠ WARNING

●For new motorcycles or when fuel has run out, do not turn on the engine stop switch before refueling. Turn it on only after adding fuel. Running the fuel pump dry will significantly reduce its service life. If the yellow malfunction indicator light illuminates during engine operation, a fault has occurred in the engine or fuel-injection system. Stop the motorcycle immediately and contact your dealer or service center for inspection.

Example: P0108 tacho signal
The fault lamp flashes 10 times in a row, then 1 time, then 10 times, and then 8 times (10 is 0)

order number	Components / systems	fault code	Fault code information	Monitoring strategy	Auxiliary parameters used for monitoring
1	1 The cylinder oxygen sensor.	P0131	1 The cylinder oxygen sensor signal is shorted to a low voltage.	Monitor the oxygen signal voltage of cylinder 1. When the oxygen voltage is below the limit value and the duration is exceeded, it is judged as a fault.	1. Cylinder oxygen sensor voltage, cylinder head / coolant temperature, and engine running time.
2		P0132	1 cylinder oxygen sensor short circuit to high voltage.	Monitor the oxygen signal voltage of cylinder 1, the oxygen voltage is above the limit and the duration is determined as a fault.	1. Cylinder oxygen sensor voltage, cylinder head / coolant temperature, and engine running time.
3		P0031	1 Cylinder oxygen sensor heating circuit short circuit to low voltage.	1 The potential on the heating output circuit is different from the ECM command.	Engine running time.
4		P0032	1 Cylinder oxygen sensor heating circuit short circuit to high voltage.	1 The potential on the heating output circuit is different from the ECM command.	Engine running time.
5	The 2-cylinder oxygen sensor.	P0137	2 cylinder oxygen sensor short circuit to low voltage.	Monitor the oxygen signal voltage of the 2 cylinders. When the rear oxygen voltage is below the limit and the duration exceeds the limit, the fault is determined as fault.	2. Cylinder oxygen sensor voltage, cylinder head / coolant temperature, and engine running time.
6		P0138	2 cylinder oxygen sensor short circuit to high voltage.	Monitor the oxygen signal voltage of cylinder 2, and when the rear oxygen voltage is above the limit and the duration exceeds the limit, it is judged as a fault.	2. Cylinder oxygen sensor voltage, cylinder head / coolant temperature, and engine running time.
7		P0037	2 cylinder oxygen sensor heating circuit short circuit to low voltage.	The potential on the heating output circuit of the 2-cylinder oxygen sensor is different from the expected potential of the ECM.	Engine running time.
8		P0038	2 cylinder oxygen sensor heating circuit short circuit to high voltage.	The potential on the heating output circuit of the 2-cylinder oxygen sensor is different from the expected potential of the ECM.	Engine running time.
9	Intake-air pressure sensor.	P0107	Intake pressure sensor line low voltage open circuit.	If the original reading and limit of intake pressure is lower, the fault is determined.	Raw reading of the air inlet pressure sensor.
10		P0108	Intake pressure sensor line high voltage.	Compare the original reading of intake pressure to the limit and determine a fault.	Raw reading of the air inlet pressure sensor.

order number	Components / systems	fault code	Fault code information	Monitoring strategy	Auxiliary parameters used for monitoring
11	Into the temperature pass sense organ.	P0112	Inlet air temperature sensor line low voltage.	Compare the original reading and limit of the incoming temperature. If it is below the limit, it is a fault.	Raw reading of the air inlet temperature sensor.
12		P0113	Intake air temperature sensor line high voltage or open circuit.	Compare the original reading and limit of the incoming temperature. If it is above the limit, it is a fault.	Raw reading of the air inlet temperature sensor.
13	Cylinder head temperature / coolant temperature, degree sensor.	P0117	Coolant temperature sensor line low voltage.	Compare the original reading to the coolant temperature sensor, and determine a fault below the limit.	Original reading of the coolant temperature sensor.
14		P0118	Coolant temperature sensor line with high voltage or open circuit.	Compare the original reading to the coolant temperature sensor, if above the limit.	Original reading of the coolant temperature sensor.
15	Fuel injector.	P0261	1-cylinder injector low voltage fault.	If the state of the cylinder injector output circuit is different from the expected state of the ECM.	
16		P0262	1-cylinder injector high voltage fault.	If the state of the cylinder injector output circuit is different from the expected state of the ECM.	
17		P0264	2-cylinder injector low voltage fault.	If the state of the 2-cylinder injector output circuit is different from the expected state of the ECM.	
18		P0265	2-cylinder injector high voltage fault.	If the state of the 2-cylinder injector output circuit is different from the expected state of the ECM.	
19	Occupy the line of fire Circle output.	P2300	1 The cylinder ignition coil is of low voltage.	If the potential on the cylinder ignition coil output circuit is different from the expected potential of the ECM, or the primary coil current is higher than the calibration setting limit, the coil is a fault.	
20		P2301	1 The cylinder ignition coil is high voltage	If the potential on the output circuit of the cylinder ignition coil is different from the expected potential of the ECM, it is determined that the coil fails	
21		P2303	2-cylinder ignition coil with low voltage.	If the potential on the output circuit of the 2-cylinder ignition coil is different from the desired potential of the ECM, or the primary coil current is above the calibration set limit, the coil fails.	
22		P2304	2-cylinder ignition coil high voltage.	If the potential on the output circuit of the 2-cylinder ignition coil is different from the expected potential of the ECM, it is determined that the coil fails.	

order number	Components / systems	fault code	Fault code information	Monitoring strategy	Auxiliary parameters used for monitoring
23	Crankshaft position sensor.	P0335	The crankshaft position sensor line has no signal.	No engine speed signal.	
24		P0336	Crankshaft position sensor line signal interference.	If the number of teeth and the 23X is read too much, it is judged as a fault.	
25	Throttle position, set the sensor.	P0122	Throttle position sensor is low voltage.	Compare the original reading and limit of throttle position sensor. If it is below the limit, it is a fault.	Original reading of the throttle position sensor.
26		P0123	Throttle position sensor high voltage.	Compare original reading and limit of throttle position sensor. If higher, fault is fault.	Original reading of the throttle position sensor.
27	Vehicle speed sensor.	P0500	Speed sensor failure.	The ECM receives a speed sensor input with an invalid speed signal.	The speed sensor input an invalid speed signal identification bit.
28	Idle speed control system.	P0505	Idle speed is abnormal.	At idle speed, the actual engine speed is above or below the set speed, and the duration exceeds the limit, then it is judged as a fault.	Engine speed, engine running time.
29	System voltage.	P0562	Low system voltage.	Comparing the system voltage with the limit value, if it is below the limit value, it is considered as a fault.	Occupation voltage.
30		P0563	High system voltage.	Comparing the system voltage to the limit, if above the limit, determine a fault.	Firing voltage
31	Oil pump after electrical equipment.	P0231	Oil pump relay open circuit / low voltage fault.	If the state of the oil pump relay output circuit during open circuit or engine operation is different from the expected state of the ECM.	
37		P0232	Oil pump relay high voltage fault.	If the condition of the oil pump relay output circuit is different from the ECM.	

order number	Components / systems	fault code	Fault code information	Monitoring strategy	Auxiliary parameters used for monitoring
38	Carbon tank electricity magnetic clamp.	P0458	Low voltage fault of carbon tank cleaning control solenoid valve.	If the state of the digging circuit of the carbon tank cleaning control solenoid valve is different from the expected state of the ECM, it is determined as a fault.	Carbon tank cleaning control solenoid valve cycle.
39		P0459	High voltage failure of carbon tank cleaning control solenoid valve.	If the state of the control circuit of the carbon tank cleaning control solenoid is different from the expected state of the ECM, it is judged as a fault.	Carbon tank cleaning control solenoid valve cycle.
40	ECM hitch.	P0601	ECM check code error.	ECM, self diagnosis.	
41	MIL.	P0650	ECM, self diagnosis	The MIL circuit status is different from the ECM output state.	The ECM output level signal.
42	Empty gear clutch, switch input.	P0850	Notral / clutch input fault.	The engine in the running state ECM monitors whether there is a shift switching action, which exceeds the limit times, it is judged as a fault.	Neutral / clutch switch signal.
43	Engine turn Fast output.	P1693	Engine speed output is low voltage.	If the state of the rotational speed output circuit is different from the desired state of the ECM, a fault is determined.	Engine speed signal.
44		P1694	Engine speed output is high voltage	If the state of the rotational speed output circuit is different from the desired state of the ECM, a fault is determined.	Engine speed signal.
45	BLM adaptability.	P0171	The system is too thin.	The system BLM learning value is greater than the limit, and the duration exceeds the limit is judged as a fault.	ELM learning values.
46		P0172	The system is too strong.	The system BLM learning value is less than the limit, and the duration exceeds the limit is judged as a fault.	BLM learning values.
47	PE pattern.	P0174	The PE mode system is too sparse.	If the FE mode is less than the limit and the duration exceeds the limit.	Oxygen signal voltage.

WARNING

●During engine operation, the fault light is not on, and the fault light flashes after turning off, which is a historical fault and has no impact on the whole vehicle, and will disappear by itself in the future.

Periodic Maintenance

Regular inspection, adjustment and lubrication can keep your locomotive in the safest and most efficient operating condition. The intervals of the periodic maintenance table should only be used as a general reference and may be shortened depending on the climate, geographical location, topography and purpose of use.

WARNING

- If the vehicle is not properly maintained, or the vehicle is maintained in an incorrect way, the risk of injury or injury may be increased during maintenance or use. If you do not know how to maintain and maintain, please entrust Xiangshuai heavy machine motorcycle dealers or service outlets.

WARNING

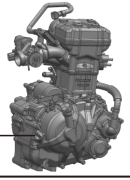
- Unless otherwise specified, please turn off the engine during locomotive maintenance. Engines that are not stalled may come into contact with the body or clothing and cause injury, and sections of the circuit may cause electric shock or fire. Performing service or maintenance work while the engine is running can cause eye injury, cause fire, or cause carbon monoxide poisoning.

WARNING

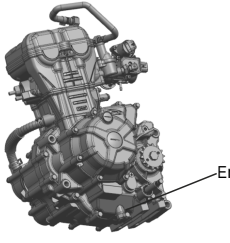
- After the motorcycle is ridden, the engine, exhaust pipe, disc brake disc, disc brake calipers, and brake pads may become hot, and in order to avoid scalding, each part can be touched after cooling.

Engine Oil Replacement

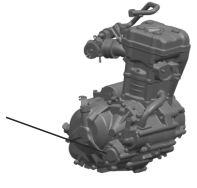
1. Insert the screw pin into the screw hole of the rear stand, then lift the motorcycle using the stand to raise the rear wheel off the ground, ensuring the vehicle remains stable.
2. Start the engine and warm it up for 2 – 3 minutes, then turn it off.
3. Place a container under the engine to collect the used engine oil.
4. Remove the engine oil filler cap, drain bolt, and washer, and drain the engine oil completely.



1. Engine oil drain bolt
2. Washer
3. The drain bolt is located at the bottom of the engine



5.Remove the oil filter cover using a special tool



NOTE:

●If engine oil replacement is not required, skip steps 5 – 7.

Oil Cap O-ring



Oil Filler Cap

Sealing Gasket



Drain Bolt

6. Place a drain pan underneath the crankcase.
7.Remove the oil filler cap, drain bolt, and sealing gasket, and drain the used engine oil completely.
8.If necessary, have the oil filter screen cleaned or replaced at an authorized service station.

Engine Oil Filling Capacity:

After engine reassembly: 1.2 L
After replacing the oil filter element: 1.0 L

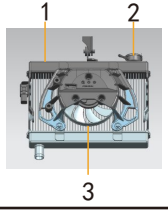
Use engine oil of grade SJ 10W-40. Be careful not to allow any foreign objects to fall into the engine crankcase during refilling.

9. Start the motorcycle and let it run at a slightly elevated speed for a few minutes. Check for any oil leakage. If leakage is found, stop the motorcycle immediately and identify the cause.

NOTE:

●Running the engine with insufficient oil will cause component damage. Excessive engine oil may affect vehicle performance and may even damage engine components.

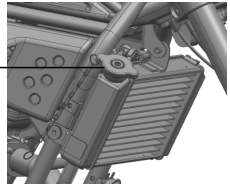
Coolant



1.Radiator
2.Radiator Cap
3.Cooling Fan

The coolant is a mixture of antifreeze concentrate and distilled water, providing both cooling and anti-freeze protection. The coolant circulates through the system to dissipate heat and reduce engine temperature. The coolant is pre-filled at the factory. During regular maintenance, check the coolant level in the reservoir. If the coolant level falls below the lower mark, add coolant as needed: open the reservoir cap and fill to the upper level mark. If the coolant inside the radiator becomes cloudy or the replacement interval is reached, replace the coolant promptly. Place the motorcycle on level ground and support it on the side stand or center stand. Remove the radiator cap and fill the radiator with coolant until it reaches the filler neck and can no longer be added. Reinstall and tighten the radiator cap. Start the engine from cold and let it idle for 2 – 3 minutes. Turn off the engine, open the reservoir cap again, and top up the coolant. Repeat this process a total of three times to fully purge air from the system. Ensure the radiator cap is tightly secured. Open the reservoir cap and add approximately 400 mL of coolant to the reservoir.

Coolant Filler Neck



CAUTION

●If coolant is not immediately available, only distilled water (free of any impurities) may be used as a temporary substitute. Do NOT use tap water, well water, mineral water, rainwater, river water, spring water, salt-containing water, or any water with impurities, as this may damage internal components of the cooling system. If distilled water is used in an emergency, replace it with proper coolant at a Marshal service center or authorized dealer as soon as possible.

●When the engine temperature reaches approximately 110°C, the temperature sensor will activate and the temperature warning indicator on the instrument panel will illuminate. This indicates that the engine must be shut down to cool. Continued operation may result in overheating and severe engine damage.

CAUTION

●Please use 3500 anti-rust and antifreeze for coolant. The coolant is toxic and corrosive, so it should never be used or contaminated on the surface of the skin or parts. The discarded liquid should be recycled and disposed of by the recycling center to avoid polluting the environment. The engine is not allowed to open the water tank cover at high temperatures to avoid injuring people from hot water spraying. If you need to disassemble the cylinder block for engine maintenance, drain the coolant first to prevent the coolant from entering the crankcase.

Cooling System Inspection

Inspect the cooling system for leaks every 400 km:

Check the water pump and all coolant hose connections for signs of leakage. Inspect coolant hoses for damage, aging, or surface cracking. Gently press the hoses by hand to check for brittleness or cracks. Replace them if any abnormalities are found. If leakage is detected, contact qualified service personnel immediately. Do not attempt to disassemble the system yourself. Check the coolant level in the reservoir. The level must remain between the upper and lower marks. Refill coolant as needed.

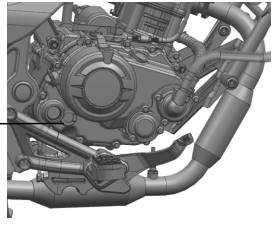
Spark Plug

Spark plugs are one of the most important components of the engine and require periodic inspection. Carbon deposits can corrode the spark plug, so regular removal, inspection, and maintenance are necessary. The condition of the spark plug also helps indicate the combustion status inside the engine ' s combustion chamber. Remove the spark plug cap and then remove the spark plug. Check whether the ceramic insulator around the center electrode is light brown in color, which indicates normal engine operation. All spark plugs installed in the engine should show similar electrode coloration. If one spark plug appears noticeably different, it may indicate abnormal combustion. If such abnormal conditions are found, have the motorcycle inspected at an authorized Jianya/Marshal Moto service center. Inspect the spark plug for any physical damage or electrode erosion. If damage is present, replace the spark plug immediately

NOTE:

●Before installing the spark plug, use a feeler gauge to measure the spark plug gap and adjust it to the specified value if necessary:
●Spark Plug Type: CPR8EA-9 (NGK)
●Spark Plug Gap: 0.8 – 0.9 mm Clean the spark plug gasket surface, the contact surface, and remove any dirt from the spark plug threads.
●Spark Plug Tightening Torque: 25 N·m. To avoid cross-threading and damaging the cylinder head threads, always start threading the spark plug by hand until it seats properly, then tighten it with a spark plug wrench. When removing the spark plug, ensure that no foreign objects fall into the engine through the spark plug opening.

Engine Oil & Oil Level Window



1.Upper Oil Level Mark
2.Engine Oil Level Inspection Window
3.Lower Oil Level Mark

●Before each ride, check the engine oil level. An oil inspection window is located at the rear of the right crankcase cover for observing the oil level. Engine oil and the oil filter must also be replaced periodically according to the maintenance schedule.

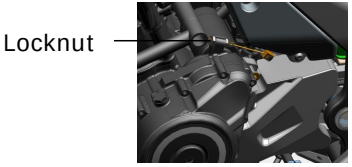
Engine Oil Level Inspection

1.Place the motorcycle on level ground and keep it upright. Any leaning may result in an incorrect oil level reading.
2.Start the engine and let it warm up for 2 – 3 minutes, then turn it off.

NOTE:

●Allow the engine oil to cool before checking the oil level to avoid inaccurate readings caused by high oil temperature.
●Let the engine sit for about 5 minutes to allow the oil to settle, then inspect the oil level through the window on the right engine cover.
●The oil level should be between the upper and lower marks on the inspection window.
●If the oil level is at or below the lower mark, add the recommended engine oil until it reaches the upper level mark.

Clutch Adjustment



Firmly hold the clutch lever and operate it.
Before obvious resistance is felt, the free play at the end of the clutch lever should be 10 – 20 mm

Brake System Inspection

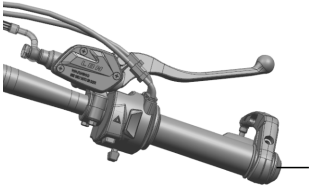
●After the brake system has been used for a certain period of time, wear will occur. When the wear reaches a certain level, please have the parts replaced at an authorized Marshalmoto service center. The minimum thickness limit of the brake disc is 3 mm.
●Check the fluid level in the brake fluid reservoir; check whether there is any fluid leakage in the brake system.
●Check the brake hose and brake fluid reservoir for leakage or cracks.
●Check the wear condition of the brake disc and brake pads.
●The front brake lever free play should be 10 – 20 mm, and the rear brake pedal free play should be 20 – 30 mm.

⚠ WARNING

●The hydraulic disc brake system operates under high pressure. To ensure safe and reliable braking, the replacement interval for brake hoses, brake fluid reservoir, and brake fluid must not exceed the time specified in this manual.

Brake System Inspection

1. Front Brake Lever

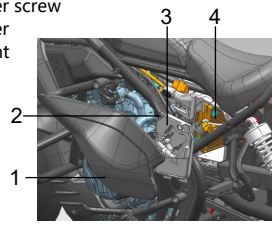


●Check the brake fluid level. The fluid level should be between the upper and lower marks. Too much or too little fluid will affect braking performance; maintain it at the proper level.

Replacement of Air Filter Element

The air cleaner should be serviced according to the regular maintenance and lubrication schedule. If the motorcycle is operated in high-humiditydusty environments, the air cleaner must be inspected and replaced more frequently.

1. Side cover
2. Air cleaner cover screw
3. Air cleaner cover
4. Air filter element



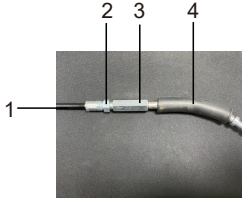
1. Remove the side cover.
2. Loosen the air cleaner cover screw.
3. Open the air cleaner cover and remove the air filter element.

NOTE：

● Do not operate the engine before reinstalling the filter element. Running the engine without a filter may cause dust to enter the engine, leading to severe piston and cylinder wear.

Throttle Cable Adjustment

1. Throttle cable
2. Locknut A
3. Adjuster nut B
4. Protective boot



1. Check whether the throttle grip rotates smoothly from fully open to fully closed.
2. Check the throttle cable between the grip and throttle body for proper function. If the cable is frayed, stuck, or routed incorrectly, it must be replaced.
3. Check the throttle free play. It should be 2 – 6 mm

If adjustment is needed:

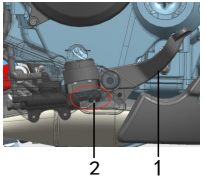
Slide the protective boot aside.Loosen locknut A.Rotate the upper adjuster nut. If further adjustment is needed, loosen adjuster nuts A and B, then adjust the adjuster screw.After achieving the correct free play (2 – 6 mm), tighten locknut A and B. Return the protective boot to its original position.

⚠ WARNING

●When operating the brake lever, if it feels softspongy, or lacks firmness, it indicates that air has entered the hydraulic brake system. In this case, do not continue riding. Please take the motorcycle to an authorized Marshalmoto service center, dealer, or service outlet to have the air removed. Air inside the brake system will significantly reduce braking effectiveness, which may lead to loss of control and serious accidents.

Rear Brake Adjustment

1. Rear brake pedal
2. Rear brake pedal adjustment bolt

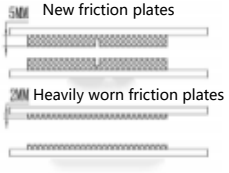


● Support the motorcycle securely and check the free play of the rear brake pedal from its resting position to the point where the brake begins to engage. If adjustment is needed: Loosen the locknut. Adjust the exposed length of the adjustment bolt to increase or decrease the pedal free play. Depress the brake pedal several times, then release it. Rotate the rear wheel to ensure it turns freely without drag. Once proper adjustment is achieved, tighten the locknut securely.

Brake Pad Inspection (Front & Rear)

Check whether the brake pads have worn close to the wear limit.

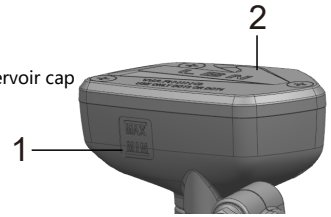
According to the Periodic Maintenance and Lubrication Schedule, regularly inspect the front and rear brake pads. The minimum allowable thickness of each brake disc must not be less than 3 mm. If the wear has exceeded the limit and is nearly touching the brake disc surface, please have the brake pads replaced at an authorized Marshalmoto dealer or service center.



Brake Fluid Level Check

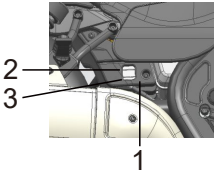
Front Brake

1. Minimum level mark
2. Front brake fluid reservoir cap



Rear Brake

1. Rear brake fluid reservoir cap
2. Maximum level mark
3. Minimum level mark



⚠ WARNING

●For hydraulic disc brakes, water or any foreign contaminants must not enter the brake fluid. Any contamination may cause serious damage to the brake system and lead to brake failure. Brake fluid is highly corrosive. Avoid spilling brake fluid onto painted surfaces or plastic parts. Use DOT 4 brake fluid only.

Inspection & Lubrication of Control Cables

Before each ride, inspect the condition of the throttle grip, throttle cable, and clutch cable. Lubricate the cables and both cable ends when necessary. If a cable is damaged or does not operate smoothly, please visit an authorized Marshalmoto dealer or service center for inspection or replacement.

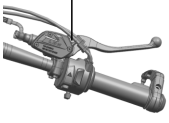
NOTE：

●A damaged cable housing can cause internal rusting, which will prevent the cable from moving smoothly. Replace any damaged cable immediately to avoid potential hazards.

Lubricate the front brake lever and clutch lever.

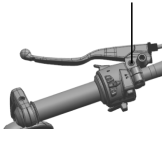
Front Brake Lever

Lubrication points

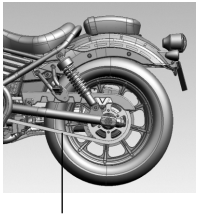


Clutch Lever

Lubrication points



Drive Chain Inspection



Drive Chain

Before each ride, inspect the drive chain length and the contact surfaces of the front and rear sprockets. Follow the maintenance manual to regularly check the condition of the chain and sprockets.

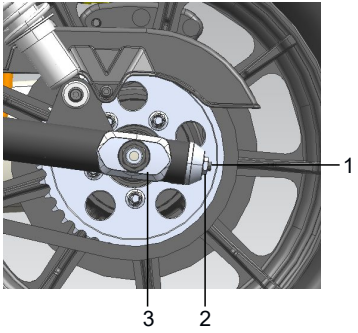
NOTE：

●If the front or rear drive sprocket has reached the wear limit, please contact an authorized Marshalmoto dealer or service center for inspection and replacement.
●Shift the transmission to neutral, and stop the engine (turn off the ignition switch).
●Use your fingers to move the chain up and down at the midpoint between the front and rear sprockets to check whether the chain slack is within the normal range.Inspect the drive chain for wear. If the chain links show missing teeth, excessive wear, or if the chain has become overly elongated, the chain must be replaced. Inspect the front and rear sprockets. If the sprocket teeth are severely worn, missing, or broken, the sprocket should be replaced immediately.

Drive Chain Adjustment & Replacement

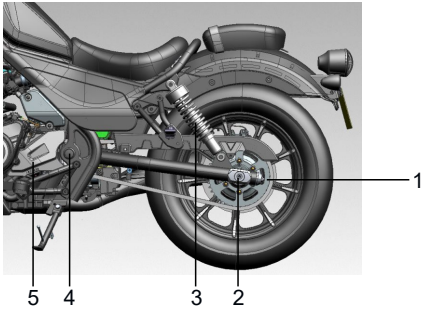
Use the center stand (or lifting stand) to raise the motorcycle so that the rear wheel is off the ground, ensuring the vehicle is stable.

1. Adjustment bolt (for M5 hex wrench)
2. Locknut (for 16 mm hex wrench)
3. Scale block



1. Shift the transmission to neutral and stop the engine (turn off the ignition switch).
2. Loosen the rear axle nut, then loosen the adjuster locknuts, and adjust the adjuster bolts by the same amount on both sides.
3. Adjust the chain to the proper slack, making sure that the left and right adjusters are set to the same position.
4. Tighten the rear axle nut and then securely tighten the locknuts.

Drive Chain Replacement



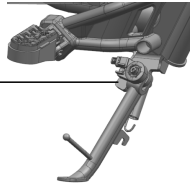
1. Loosen the chain adjuster nuts and remove the left and right chain adjusters.
2. Remove the rear axle and take off the rear wheel.Remove the left and right rear shock absorbers.
3. Remove the swingarm pivot shaft and take off the rear swingarm.
4. Remove the chain cover.

NOTE：

●After reassembly, ensure that the left and right adjusters are evenly aligned, and tighten all locknuts securely.

Side Stand Inspection & Lubrication

Lubrication of side brackets



WARNING

●If the single brace cannot rise and fall smoothly, please go to the Xiangshuai Heavy Machinery dealer or service outlet for maintenance, otherwise the single brace may touch the ground while driving, causing the driver to lose control due to distraction.

Shock absorption before inspection

●Check the condition of the fork regularly in accordance with the provisions of the maintenance manual.
●Please check whether there are scratches, wear and oil leakage on the surface of the shock absorbing cylinder core.

1. Park your car on flat ground and keep your vehicle upright.
2. Grasp the front brake and press down firmly on the handlebar several times. Check that the front shock absorbers have smooth compression and rebound.

NOTE：

●If any damage is found or the front shock absorption cannot be operated smoothly, please contact the dealer or service point of Xiangshuai Heavy Machinery for inspection or replacement.

Direction column check

Please check the steering column regularly according to the maintenance regulations in this manual, as worn or loose directional column bearings may cause dangerous conditions.

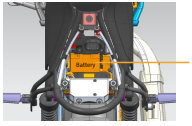
1. Insert the screw into the rear flat fork shaft hole, and then use the landing gear to lift the car so that the rear wheels leave the ground to ensure that the car is stable.
2. Grasp the lower part of the front shock absorber and try to shake it forward and back. If you can feel a gap, please contact the Guangjian Ya motorcycle dealer or service point for inspection and maintenance.

Front & Rear Wheel Bearing Inspection

Regularly check the vehicle bearings for wear in accordance with the maintenance manual. If there is cross-clearance between the left and right sides of the wheel or the wheel cannot rotate smoothly, please contact Xiangshuai Heavy Machinery motorcycle dealer or service outlet for inspection and maintenance.

Battery Inspection

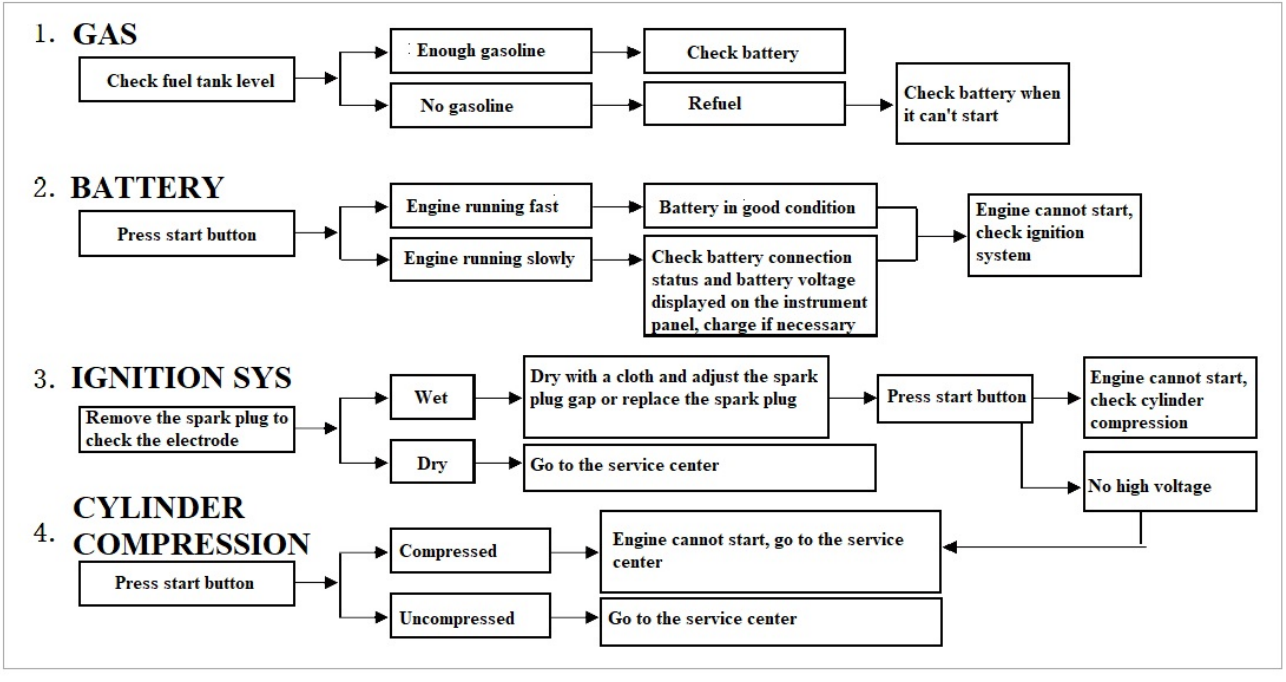
1. Negative power cord "black"
2. Positive power cord "red"



The battery is installed under the driver's seat. The machine car is equipped with a lead-acid battery, there is no need to check the electrolyte or replenish the distilled water. However, it is necessary to check the wire connection of the battery, and if necessary, check whether the battery screw is locked.

Troubleshooting Chart

Starting problems or engine problems



⚠ WARNING

●Battery electrolyte contains sulfuric acid, which is toxic and dangerous, causing severe burns, so avoid contact with skin, eyes , or clothing. When working near batteries, bring goggles. If you accidentally come into contact, please take the following first aid measures:
●Accidental drinking: Drink a lot of water or milk in the body and send it to the hospital immediately.

●Accidental entry into eyes: Rinse immediately with water for 15 minutes and immediately send to the hospital.
●The battery will produce explosive gases, please keep away from sparks, flames, cigarettes and other fire sources, and maintain sufficient ventilation when charging in a closed space. The battery will produce explosive gases, please keep away from sparks, flames, cigarettes and other ignition sources, and maintain sufficient ventilation when charging in an enclosed space.
●Keep the battery out of the reach of children.

Battery Charging

If you feel that the battery is insufficient, check the voltage across the battery, such as 12.3V or less, please charge it immediately, if the vehicle has other optional electrical accessories, so that the battery discharge faster.

Storage of batteries

1. If the locomotive is not used for more than 1 month, please remove the battery and store it in a cool and dry place.

NOTE:

● When removing the battery, remember to turn off the power of the vehicle first, and the order of removing the power cord is: first remove the negative wire, and then remove the positive wire.

NOTE:

2. If the battery is to be stored for more than two months, check it at least once a month, and fully charge the battery if necessary.

●When installing the battery, connect the negative wire first, and then connect the positive wire.

3. After installation, confirm that the battery wires are connected correctly.

NOTE:

When installing and removing the battery, connect the negative wire first, and then connect the positive wire.

Fuse Replacement

The cable fuse box, which contains fuses for other circuits, is located under the seat cushion of the vehicle, and if the fuse burns, replace the fuse as follows.

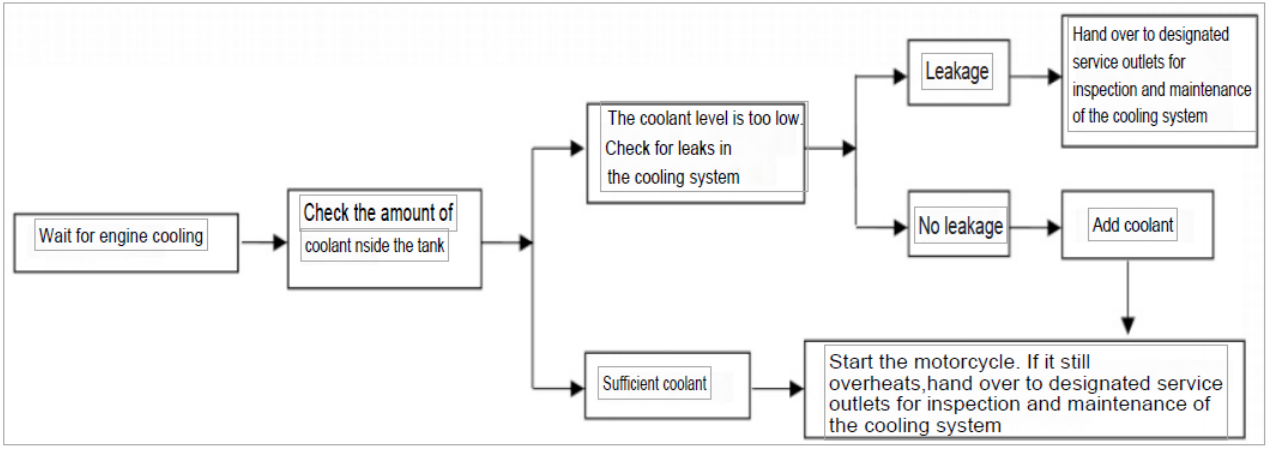
1. Turn off the power of the vehicle and open the seat cushion.
2. Remove the burnt fuse and install a new fuse with the specified amperage value.

⚠ WARNING

●Do not replace the fuse larger than the specified number value to avoid damage to the electrical system and cause fire.

Engine Overheating

When the engine water tank is hot.Do not remove the cover of the water tank.Hot coolant and steam can spray out under pressure,causingserious scalding accidents Be sure to wait until theengine has cooled down beforeproceeding.Cover the lid of the tank with a thick cloth such as a towel,slowly screw the cover to the edge of the card slot in acounterclockwise direction to release the remaining pressure in the tank,press the lid down when it is finished,and then rotate it counterdockwise to remove.



Maintenance cycle table

Order number	Maintenance project	Kilometer number of the maintenance cycle							Remarks
		1000KM	4000KM	7000KM	10000KM	13000KM	16000KM	19000KM	
13	Storage battery	check up	The examination of oneself	check up	check up	check up	check up	check up	Do not use for a long time, should be charged once a month
14	Brake lining	The examination of oneself	The examination of oneself	check up	check up	check up	check up	check up	The thickness of the friction area shall not be less than 2MM
* 15	Brake assembly	Check the interface	check up	check up	check up	check up	The examination of oneself	check up	Fill the brake fluid if necessary
16	The brake light switch	The examination of oneself	Check whether	check up	check up	check up	check up	check up	
17	Lighting system	The examination of oneself	check up	check up	Check yao	The examination of oneself	The examination of oneself	check up	
* 18	Clutch	The examination of oneself	The examination of oneself	The examination of oneself	check up	The examination of oneself	The examination of oneself	The examination of oneself	
* 19	Shock absorber	check up	check up	The examination of oneself	check up	check up	check up	check up	
20	Bolt / nut	adjust	adjust	adjust	adjust	adjust	adjust	adjust	Critical bolt adjustment torque
21	Front and rear tires	check up	check up	check up	check up	check up	check up	check up	
22	Direction-pressure bearing	Impediment / adjustment	Check / adjust	Impediment / adjustment	Check / adjust	Check / adjust	Impediment / adjustment	Impediment / adjustment	Adjust by professionals when necessary
23	Coolant liquid	Impediment / adjustment	Impediment / adjustment	Impediment / adjustment	Check / adjust	Check / adjust	Impediment / adjustment	Check / adjust	

Note: When driving in areas with abnormally wet or dusty areas, the maintenance interval should be appropriately shortened. Items marked with * can only be adjusted by professionals.

Maintenance cycle table

Order number	Maintenance project	Kilometer number of the maintenance cycle							Remarks
		1000KM	4000KM	7000KM	10000KM	13000KM	16000KM	19000KM	
13	Storage battery	check up	The examination of oneself	check up	check up	check up	check up	check up	Do not use for a long time, should be charged once a month
14	Brake lining	The examination of oneself	The examination of oneself	check up	check up	check up	check up	check up	The thickness of the friction area shall not be less than 2MM
* 15	Brake assembly	Check the interface	check up	check up	check up	check up	The examination of oneself	check up	Fill the brake fluid if necessary
16	The brake light switch	The examination of oneself	Check whether	check up	check up	check up	check up	check up	
17	Lighting system	The examination of oneself	check up	check up	Check yao	The examination of oneself	The examination of oneself	check up	
* 18	Clutch	The examination of oneself	The examination of oneself	The examination of oneself	check up	The examination of oneself	The examination of oneself	The examination of oneself	
* 19	Shock absorber	check up	check up	The examination of oneself	check up	check up	check up	check up	
20	Bolt / nut	adjust	adjust	adjust	adjust	adjust	adjust	adjust	Critical bolt adjustment torque
21	Front and rear tires	check up	check up	check up	check up	check up	check up	check up	
22	Direction-pressure bearing	Impediment / adjustment	Check / adjust	Impediment / adjustment	Check / adjust	Check / adjust	Impediment / adjustment	Impediment / adjustment	Adjust by professionals when necessary
23	Coolant liquid	Impediment / adjustment	Impediment / adjustment	Impediment / adjustment	Check / adjust	Check / adjust	Impediment / adjustment	Check / adjust	

Note: When driving in areas with abnormally wet or dusty areas, the maintenance interval should be appropriately shortened. Items marked with * can only be adjusted by professionals.

Headlamps

This machine is equipped with LED headlights, if the headlights do not light up, please go to Xiangshuai heavy machinery motorcycle dealers or service outlets to check the electrical wiring.

NOTE：

●Do not attach any colored film to the headlamp lens.

Front position light

This machine is equipped with LED front position light, if the front position light is not on, please go to Xiangshuai heavy machine motorcycle dealer or service outlet to check the electrical wiring.

Taillight/brake light replacement

This machine car is equipped with LED integrated brake light tail light, if the brake light tail light is not on, please go to Xiangshuai heavy machinery motorcycle dealer or service outlet to check the electrical wiring.

Replacement of front turn signals

This machine is equipped with LED front turn signals, if the turn signals do not turn on, please go to Xiangshuai heavy machinery motorcycle dealers or service outlets to check the electrical wiring.

Care

Even with the highest quality parts, it is easy to rust and corrode. Regular and correct care not only meets the requirements of maintenance, but also makes your locomotive look beautiful, extend its service life and give full play to its performance.

1.After the engine cools, put a plastic bag on the tail of the exhaust muffler.
2.Make sure all covers, plugs, and all electrical connectors and terminals (including spark plug caps) are properly installed.
3.Remove dirt.

NOTE：

● Do not use strong acidic detergent, if you use such cleaning products, do not let the detergent remain for too long. Also, immediately dry with a soft towel after rinsing with clean water and spray with anti-rust spray.
● Improper cleaning can damage shrouds, panels, windshields, headlamp shades, odometer covers, and other plastic parts. Only use a soft cloth and sponge dampened with water. If there is dirt on the plastic that cannot be removed with water, the neutral detergent can be diluted with water. Please rinse the remaining detergent with clean water to avoid damaging the plastic parts.
● Do not use high-pressure car washers or steam jet cleaners for cleaning, as they can cause water leakage and deterioration of the following parts: sealing parts (wheel bearings, forks, brake and fork bearings), electrical components (meters, connector terminals, switches, lamps), ventilation hoses, ventilation ducts , and cables.

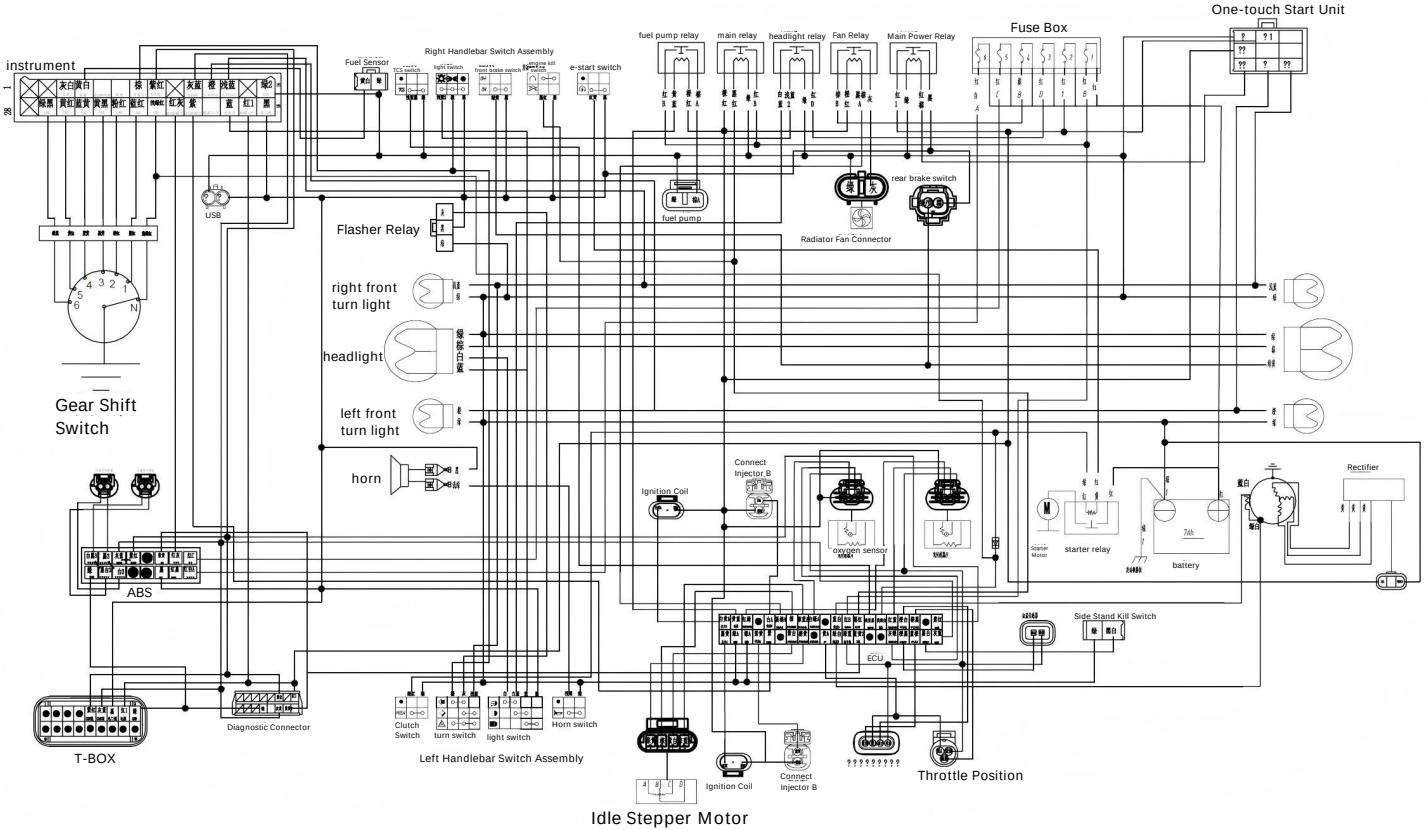
Storage

Store for a short time

Please store the locomotive in a dry and cool place; Cover with a dust cover if necessary to keep out dust, and make sure the engine and exhaust muffler have cooled before removing the cover.

It needs to be stored for a long time for several months

1. Fill the gasoline tank with gasoline and add gasoline stabilizer (if any) to avoid rust and gasoline deterioration in the tank.
2. In order to avoid rust of cylinders, piston rings, etc., please follow the following requirements. (Remove the spark plug, such as 15ml of engine oil into the spark plug hole inside the point.)
3. Check the tire pressure and adjust it if necessary. Rotate the wheels slightly every month to prevent the tire from deteriorating easily due to fixing a little.
4. Wrap the muffler mouth with a plastic bag to avoid moisture entering.
5. Remove the battery and store it in a cool and dry place, and charge it once a month.


One-touch Start Unit
Fuse Box
fuel pump relay
main relay
headlight relay
Fan Relay
Main Power Relay
instrument
Fuel Sensor
Right Handlebar Switch Assembly
e-start switch
USB
fuel pump
rear brake switch
Radiator Fan Connector
Flasher Relay
right front turn light
headlight
left front turn light
horn
Gear Shift Switch
Ignition Coil
Connect Injector B
oxygen sensor
starter relay
battery
Rectifier
ABS
ECU
Side Stand Kill Switch
T-BOX
Diagnostic Connector
Clutch Switch
turn switch
light switch
Horn switch
Left Handlebar Switch Assembly
Ignition Coil
Connect Injector B
Throttle Position
Idle Stepper Motor

Vehicle dimensions
Full length: 2200mm
Full width: 815mm
Full height: 1160mm
Wheelbase: 1500mm
Minimum ground clearance: 186mm

weight
Curb weight: 153KG
engine
Form: Four-stroke, DOHC
Cooling method: water cooling

Number of cylinders: single cylinder
Displacement: 124.18ml
Bore*Stroke: 58 * 47
Compression ratio: 11.5:1
Start mode: electric start

Engine oil
Grade: 10W-40
Grade: API/SG or higher Coolant

capacity
Water tank capacity: 1000ml

Air filter
Air filter element: paper element filter element gasoline

Recommended gasoline: unleaded gasoline above 92#
capacity: 14L

Spark plug
manufacturer/Model: CPR8EA-9 (NGK)
Spark plug gap: 0.8-0.9mm

Clutch type
Wet, multi-piece, manual transmission
Primary drive ratio: 2.929
Final drive: chain
Gear type: 6 gears / international gear

Frame
Frame form: steel tube braided frame
Front angle: 29°

Front tires
Tire type: high-speed tubeless
Tire model/specification: 120/80-16
Rear tires
Tire type: high-speed tubeless
Tire model/specification: 150/80-16

Maximum load capacity
Load limit: 323KG
*(Total quantity limit for riders , passengers, cargo and accessories)

Tire pressure
Front wheels: 250Kpa
Rear: 250Kpa
Front hub
Hub form: aluminum alloy casting
Wheel specifications: 3.0*16
Rear hub
Hub form: aluminum alloy casting
Wheel specifications: 3.5*16
Front brake
Type: Single caliper hydraulic disc type
Operation mode: right handle
Specified brake fluid: DOT4
Rear braking
Type: Single caliper hydraulic disc type

Mode of operation: right foot brake pedal
Specified brake fluid: DOT4 Front suspension
Type: Front minus upright
Shock absorber type: hydraulic shock absorber Rear suspension
Type: Upright

Shock absorber type: spring/oil pressure
Electrical installations Ignition system: Ignition ignition system

Charging system: AC engine accumulator
Model: 10HR6MF27S
Voltage and capacity: 12V8Ah

Lamp voltage, watts and quantity:
Headlamps: LED, DC12V
High beam 30W Low beam 16W
Tail light/brake light: LED, DC12V 1.65W/2.5W

Front turn signals: LED, DC12V 1W
Rear turn signals: LED, DC12V 1W
Position light: LED, DC12V 0.5W
license plate light: LED, DC12V 1W
Dashboard light: LED
High beam indicator: LED, DC12V
Engine failure warning light: LCD

ABS warning light: LCD Fuse:
Total fuse: 25A
Lighting fuse: 7.5A
EFI fuse: 25A
ABS fuse: 15A

MITT

Maintenance Schedule

JOY125 | Maintenance and Adjustment

No.	Maintenance Item	1000 km	4000 km	7000 km	10000 km	13000 km	16000 km	19000 km	Remarks
1	Fuel system	Inspect	Inspect	Inspect	Inspect	Inspect	Inspect	Inspect	Adjust when necessary
2	Control cables	Inspect	Inspect/Adjust	Inspect/Adjust	Inspect/Adjust	Inspect/Adjust	Inspect/Adjust	Inspect/Adjust	
*3	Fuel injection system	Inspect/Adjust	Inspect/Adjust	Inspect/Adjust	Inspect/Adjust	Inspect/Adjust	Inspect/Adjust	Inspect/Adjust	Adjusted by professional personnel
4	Fuel filter	Inspect	Inspect	Inspect	Inspect	Inspect	Inspect	Inspect	Replace with original parts when necessary
5	Air filter element	Adjust	Adjust	Adjust	Adjust	Adjust	Adjust	Adjust	Replace with original parts when necessary
6	Spark plug gap	Inspect/Adjust	Inspect/Adjust	Inspect/Adjust	Inspect/Adjust	Inspect/Adjust	Inspect/Adjust	Inspect/Adjust	Adjusted by professional personnel
*7	Valve clearance	Inspect	Inspect	Inspect	Inspect	Inspect	Inspect	Inspect	Adjusted by professional personnel
8	Engine oil	Replace	Replace	Replace	Replace	Replace	Replace	Replace	
*9	Oil filter	Replace	Replace	Replace	Replace	Replace	Replace	Replace	
*10	Timing chain	Inspect	Inspect	Inspect	Inspect	Inspect	Inspect	Inspect	Adjust when necessary
*11	Idle speed	Inspect	Inspect	Inspect	Inspect	Inspect	Inspect	Inspect	Adjust when necessary
12	Drive belt	Inspect every 2000 km							
13	Battery	Inspect	Inspect	Inspect	Inspect	Inspect	Inspect	Inspect	If not used for a long time, charge once a month
14	Brake pads	Inspect	Inspect	Inspect	Inspect	Inspect	Inspect	Inspect	Friction area thickness must not be less than 2 mm
*15	Brake system	Inspect	Inspect	Inspect	Inspect	Inspect	Inspect	Inspect	Top up brake fluid when necessary
16	Brake light switch	Inspect	Inspect	Inspect	Inspect	Inspect	Inspect	Inspect	
17	Lighting system	Inspect	Inspect	Inspect	Inspect	Inspect	Inspect	Inspect	
*18	Clutch	Inspect	Inspect	Inspect	Inspect	Inspect	Inspect	Inspect	
*19	Shock absorbers	Inspect	Inspect	Inspect	Inspect	Inspect	Inspect	Inspect	
20	Bolts / Nuts	Adjust	Adjust	Adjust	Adjust	Adjust	Adjust	Adjust	Key bolts: check tightening torque
21	Front and rear tires	Inspect	Inspect	Inspect	Inspect	Inspect	Inspect	Inspect	
22	Steering bearings	Inspect/Adjust	Inspect/Adjust	Inspect/Adjust	Inspect/Adjust	Inspect/Adjust	Inspect/Adjust	Inspect/Adjust	Adjusted by professional personnel when necessary
23	Coolant	Inspect/Adjust	Inspect/Adjust	Inspect/Adjust	Inspect/Adjust	Inspect/Adjust	Inspect/Adjust	Inspect/Adjust	

Note: When riding in areas with abnormal humidity or high dust levels, the maintenance intervals should be appropriately shortened.
*Items marked with * can only be adjusted by professional personnel.*

Legend: Inspect = Check condition | Inspect/Adjust = Check and adjust as needed | Replace = Replace with new part | Adjust = Adjust to specification

JOY125 OBD Fault Code List

Diagnostic Trouble Codes (DTC) – Components, Descriptions & Monitoring Strategies

No.	Component / System	Fault Code	Fault Description	Monitoring Strategy	Auxiliary Parameters
1	Cylinder 1 Oxygen Sensor	P0131	Cylinder 1 oxygen sensor signal shorted to low voltage	Monitor oxygen signal voltage of cylinder 1. If voltage is below the limit and duration is exceeded, a fault is determined.	Cylinder 1 O2 sensor voltage, cylinder head/coolant temperature, engine running time
2	Cylinder 1 Oxygen Sensor	P0132	Cylinder 1 oxygen sensor signal shorted to high voltage	Monitor oxygen signal voltage of cylinder 1. If voltage is above the limit and duration is exceeded, a fault is determined.	Cylinder 1 O2 sensor voltage, cylinder head/coolant temperature, engine running time
3	Cylinder 1 Oxygen Sensor Heater	P0031	Cylinder 1 oxygen sensor heater circuit shorted to low voltage	Potential on the heater output circuit differs from the ECM command.	Engine running time
4	Cylinder 1 Oxygen Sensor Heater	P0032	Cylinder 1 oxygen sensor heater circuit shorted to high voltage	Potential on the heater output circuit differs from the ECM command.	Engine running time
5	Cylinder 2 Oxygen Sensor	P0137	Cylinder 2 oxygen sensor signal shorted to low voltage	Monitor oxygen signal voltage of cylinder 2. If voltage is below the limit and duration is exceeded, a fault is determined.	Cylinder 2 O2 sensor voltage, cylinder head/coolant temperature, engine running time
6	Cylinder 2 Oxygen Sensor	P0138	Cylinder 2 oxygen sensor signal shorted to high voltage	Monitor oxygen signal voltage of cylinder 2. If voltage is above the limit and duration is exceeded, a fault is determined.	Cylinder 2 O2 sensor voltage, cylinder head/coolant temperature, engine running time
7	Cylinder 2 Oxygen Sensor Heater	P0037	Cylinder 2 oxygen sensor heater circuit shorted to low voltage	Potential on the heater output circuit of cylinder 2 O2 sensor differs from the expected ECM potential.	Engine running time
8	Cylinder 2 Oxygen Sensor Heater	P0038	Cylinder 2 oxygen sensor heater circuit shorted to high voltage	Potential on the heater output circuit of cylinder 2 O2 sensor differs from the expected ECM potential.	Engine running time
9	Intake Air Pressure Sensor	P0107	Intake pressure sensor circuit low voltage / open	If the raw reading of intake pressure is below the limit, a fault is determined.	Raw reading of the intake air pressure sensor
10	Intake Air Pressure Sensor	P0108	Intake pressure sensor circuit high voltage	Compare the raw reading of intake pressure to the limit. If above the limit, a fault is determined.	Raw reading of the intake air pressure sensor
11	Intake Air Temperature Sensor	P0112	Intake air temperature sensor circuit low voltage	Compare the raw reading of intake air temperature to the limit. If below the limit, a fault is determined.	Raw reading of the intake air temperature sensor
12	Intake Air Temperature Sensor	P0113	Intake air temperature sensor circuit high voltage / open	Compare the raw reading of intake air temperature to the limit. If above the limit, a fault is determined.	Raw reading of the intake air temperature sensor
13	Coolant Temperature Sensor	P0117	Coolant temperature sensor circuit low voltage	Compare the raw reading of the coolant temperature sensor to the limit. If below the limit, a fault is determined.	Raw reading of the coolant temperature sensor
14	Coolant Temperature Sensor	P0118	Coolant temperature sensor circuit high voltage / open	Compare the raw reading of the coolant temperature sensor to the limit. If above the limit, a fault is determined.	Raw reading of the coolant temperature sensor
15	Cylinder 1 Injector	P0261	Cylinder 1 injector circuit low voltage	State of the cylinder 1 injector output circuit differs from the expected state of the ECM.	
16	Cylinder 1 Injector	P0262	Cylinder 1 injector circuit high voltage	State of the cylinder 1 injector output circuit differs from the expected state of the ECM.	

JOY125 OBD Fault Code List

Diagnostic Trouble Codes (DTC) – Components, Descriptions & Monitoring Strategies

No.	Component / System	Fault Code	Fault Description	Monitoring Strategy	Auxiliary Parameters
17	Cylinder 2 Injector	P0264	Cylinder 2 injector circuit low voltage	State of the cylinder 2 injector output circuit differs from the expected state of the ECM.	
18	Cylinder 2 Injector	P0265	Cylinder 2 injector circuit high voltage	State of the cylinder 2 injector output circuit differs from the expected state of the ECM.	
19	Cylinder 1 Ignition Coil	P2300	Cylinder 1 ignition coil circuit low voltage	Potential on the cylinder 1 ignition coil output circuit differs from the expected potential of the ECM.	
20	Cylinder 1 Ignition Coil	P2301	Cylinder 1 ignition coil circuit high voltage	Potential on the cylinder 1 ignition coil output circuit differs from the expected potential of the ECM.	
21	Cylinder 2 Ignition Coil	P2303	Cylinder 2 ignition coil circuit low voltage	Potential on the cylinder 2 ignition coil output circuit differs from the expected potential of the ECM.	
22	Cylinder 2 Ignition Coil	P2304	Cylinder 2 ignition coil circuit high voltage	Potential on the cylinder 2 ignition coil output circuit differs from the expected potential of the ECM.	
23	Crankshaft Position Sensor	P0335	Crankshaft position sensor circuit – no signal	No engine speed signal detected.	
24	Crankshaft Position Sensor	P0336	Crankshaft position sensor circuit – signal interference	Number of teeth detected differs significantly from expected (23X). Fault is determined if excessive.	
25	Throttle Position Sensor	P0122	Throttle position sensor circuit low voltage	Compare the raw reading of the throttle position sensor to the limit. If below the limit, a fault is determined.	Raw reading of the throttle position sensor
26	Throttle Position Sensor	P0123	Throttle position sensor circuit high voltage	Compare the raw reading of the throttle position sensor to the limit. If above the limit, a fault is determined.	Raw reading of the throttle position sensor
27	Vehicle Speed Sensor	P0500	Vehicle speed sensor failure	ECM receives an invalid speed signal from the speed sensor input.	Invalid speed signal identification bit from speed sensor
28	Idle Speed Control System	P0505	Idle speed abnormal	At idle, actual engine speed is lower or higher than the estimated speed by more than the limit. Fault is determined if duration is exceeded.	Engine speed, engine running time
29	System Voltage	P0562	System voltage low	Compare system voltage to the limit. If below the limit, a fault is determined.	System / battery voltage
30	System Voltage	P0563	System voltage high	Compare system voltage to the limit. If above the limit, a fault is determined.	System / battery voltage
31	Fuel Pump Relay	P0231	Fuel pump relay circuit open / low voltage	State of the fuel pump relay output circuit (open or during engine operation) differs from the expected state of the ECM.	
37	Fuel Pump Relay	P0232	Fuel pump relay circuit high voltage	State of the fuel pump relay output circuit differs from the expected state of the ECM.	

JOY125 OBD Fault Code List

Diagnostic Trouble Codes (DTC) – Components, Descriptions & Monitoring Strategies

No.	Component / System	Fault Code	Fault Description	Monitoring Strategy	Auxiliary Parameters
38	Canister Purge Solenoid	P0458	Canister purge control solenoid valve circuit low voltage	State of the canister purge control solenoid circuit differs from the expected state of the ECM.	Canister purge control solenoid duty cycle
39	Canister Purge Solenoid	P0459	Canister purge control solenoid valve circuit high voltage	State of the canister purge control solenoid circuit differs from the expected state of the ECM.	Canister purge control solenoid duty cycle
40	ECM	P0601	ECM checksum / internal error	ECM self-diagnosis detects checksum error.	
41	MIL (Malfunction Indicator Lamp)	P0650	MIL circuit malfunction	MIL circuit status differs from the ECM output state.	ECM output level signal
42	Neutral / Clutch Switch	P0850	Neutral / clutch input circuit fault	While engine is running, ECM monitors neutral switch input. If signal is missing or exceeds limit times, a fault is determined.	Neutral / clutch switch signal
43	Engine Speed Output	P1693	Engine speed output circuit low voltage	State of the engine speed output circuit differs from the desired state of the ECM.	Engine speed signal
44	Engine Speed Output	P1694	Engine speed output circuit high voltage	State of the engine speed output circuit differs from the desired state of the ECM.	Engine speed signal
45	Fuel System (BLM)	P0171	System too lean (BLM adaptation)	BLM learning value is greater than the upper limit and duration is exceeded. Fault is determined.	BLM learning values
46	Fuel System (BLM)	P0172	System too rich (BLM adaptation)	BLM learning value is less than the lower limit and duration is exceeded. Fault is determined.	BLM learning values
47	PE Mode (Power Enrichment)	P0174	PE mode system too lean	PE mode signal is below the limit and duration is exceeded. Fault is determined.	PE mode signal voltage

Notes:

1. Original document OCR quality was poor. Descriptions and strategies have been cleaned and standardized into clear technical English while preserving original meaning.

2. Order numbers follow the source document (some numbers are missing, e.g. 32–36).

3. P0458 / P0459: Original showed 'P0450' and 'P045V' – corrected to standard canister purge codes based on context.

4. BLM = Block Learn Multiplier (long-term fuel trim adaptation). PE = Power Enrichment mode.

5. This table is for reference / service documentation of the JOY125 model.